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Discovering human germ cell mutagens with whole genome sequencing: Insights from power calculations reveal the importance of controlling for between-family variability

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ABSTRACT

Mutations in germ cells pose potential genetic risks to offspring. However, *de novo* mutations are rare events that are spread across the genome and are difficult to detect. Thus, studies in this area have generally been underpowered, and no human germ cell mutagen has been identified. Whole Genome Sequencing (WGS) of human pedigrees has been proposed as an approach to overcome these technical and statistical challenges. WGS enables analysis of a much wider breadth of the genome than traditional approaches. Here, we performed power analyses to determine the feasibility of using WGS in human families to identify germ cell mutagens. Different statistical models were compared in the power analyses (ANOVA and multiple regression for one-child families, and mixed effect model sampling between two to four siblings per family). Assumptions were made based on parameters from the existing literature, such as the mutation-by-paternal age effect. We explored two scenarios: a constant effect due to an exposure that occurred in the past, and an accumulating effect where the exposure is continuing. Our analysis revealed the importance of modeling inter-family variability of the mutation-by-paternal age effect. Statistical power was improved by models accounting for the family-to-family variability. Our power analyses suggest that sufficient statistical power can be attained with 4–28 four-sibling families per treatment group, when the increase in mutations ranges from 40 to 10% respectively. Modeling family variability using mixed effect models provided a reduction in sample size compared to a multiple regression approach. Much larger sample sizes were required to detect an interaction effect between environmental exposures and paternal age. These findings inform study design and statistical modeling approaches to improve power and reduce sequencing costs for future studies in this area.

1. Introduction

The Human Gene Mutation Database contains over 200,000 genetic alterations that cause inherited disease [1]. Mutations within this database include single nucleotide variants (SNVs), insertions/deletions (indels) of different sizes, copy number variants, translocations, and complex rearrangements. The majority of the events in the database are SNVs and indels, and originated as *de novo* mutations (DNM). DNM occur in every individual and affect the prevalence of genetic disorders in human populations. Offspring with DNM are at higher risk of genetic diseases [2–6] and there is a fast growing catalogue of genomic sites associated with human genetic disorders [1,7,8]. For example, between 1 in 200–500 offspring are born with DNM that have an association with intellectual developmental disorders [9]. An elusive question in this emerging area of research is to what extent are germline mutation rates affected by environmental and lifestyle factors.

One established variable that is associated with increased incidence of DNM in humans is paternal age. For every additional year of paternal age at conception, fathers transmit additional DNM to their offspring [10–13]. Intriguingly, one study applying whole genome sequencing to three multi-sibling families showed that there is large variability in germline mutation rates between fathers [10]. This suggests that, as well as paternal age at conception of the child and variability in DNA repair phenotypes, there may be exogenous factors that contribute to germline mutation rates. One possibility is that fathers accrue additional germ cell mutations through exposure to environmental mutagens (*e.g.*, cigarette smoke [14], radiation [15–17], air pollution [18], *etc.*).

A variety of methods have been developed to study germline mutagenesis in rodent models [19]. Methods to measure heritable genetic effects have generally required large sample sizes, with analyses focussed on a small portion of the genome; for example, analysis of

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induced mutations using the mouse specific locus test [20]. Studies applying these methods have shown that a wide range of agents in our environment are capable of causing DNM in animals [21]. However, despite the similarities in mutagenesis and spermatogenesis between humans and rodents, there has been limited evidence of human germ cell mutagens.

The study of environmentally induced human germline mutations has been historically limited by a lack of suitable tools to identify mutations, leading to insufficient statistical power. Heritable mutations are rare events, generally with less than one hundred DNM per person spread throughout a three billion base-pair genome [11]. Human studies to identify possible environmental exposures that induce DNM have traditionally measured phenotypic endpoints; as only a small fraction of DNM result in a loss-of-function mutation, the effect size is markedly reduced, leading to underpowered tests [21]. Human observational studies are impacted by many sources of noise including variability in human genetics, lifestyles, and environmental exposures [16,22–24], which further reduce power. To address these important challenges, it has been proposed that Whole Genome Sequencing (WGS) be used to detect human germ cell mutagens [25], but this has not been applied in practice yet.

Combining genomic and epidemiology data provides a new opportunity to detect human germ cell mutagens, while mitigating the limitations of epidemiology data. WGS allows the inspection of a greater breadth of the genome, providing a more accurate estimate of the heritable mutations transmitted to offspring. This greater accuracy from whole genome analysis achieves sufficient statistical power with smaller, more feasible sample sizes [25–27] and reveals information about the mutational mechanisms [28,29]. In effect, this genomic epidemiology approach flips the experimental paradigm from high sample size and low genome breadth to modest sample size and high genome breadth. In addition, WGS can provide mechanistic predictions to detect differences in mutational spectra between control and exposed groups due to the distinct nucleotide substitution patterns caused by mutagens [30,31], increasing the confidence that an exposure contributes to heritable mutations [32]. For example, the mutational signature of cigarette smoke is characterised by C > T transitions and C > A transversions [33]; therefore, it would be expected that C > T and C > A mutations will be more frequent in the offspring of fathers who smoked. Given the promise of genomic approaches in identifying environmental mutagens, it is critical to consider the feasibility of applying WGS in humans to detect induced DNM.

The objective of this study was to determine the best study design and statistical model to maximise power when detecting environmentally induced mutations in the human germline. This involved performing a power analysis, informed from existing whole genome human pedigree data, across a range of realistic effect sizes. Two scenarios were considered. The first scenario is one in which the exposure occurred in the past. In this scenario there is a fixed number of environmentally induced DNM that are transmitted to offspring on top of the paternal age mutation effect. We called this scenario the 'constant effect scenario.' In the second scenario, there is an accumulating effect of the environmental exposure (*i.e.*, an increasing number of DNM with each year of the exposure) on top of the paternal age effect. We called this scenario the 'accumulating effect' scenario. An example of the constant effect scenario is a father who smoked, but stopped smoking prior to the conception of children, where a fixed number of mutations occurred in his stem cell population. An example of the accumulating effect scenario is a father who was an active cigarette smoker over the period of time in which his children were conceived, where the number of mutations was continually increasing over time in his stem cells as a result of the ongoing exposure. Our goal was to establish a practical approach by which WGS can be applied to detect human germ cell mutagens and make recommendations about sample allocation and statistical models.

2. Methods

We used a simulation approach to conduct power analyses for detecting environmentally induced increases in DNM using established methods [34,35]. Our modeling accounted for random effects (*e.g.*, each father responds to the mutagen differently as he ages, providing a source of random variation) [10]. We note that there is hierarchical structure for these data (*e.g.*, siblings are nested within families to establish mutation-by-paternal age slopes), which influences the type of statistical model that can be conducted.

Data were simulated for several important biological variables across different scenarios. Variables identified to be important for simulating the rates of DNM were: paternal age when the child was conceived, the variability of the mutation-by-paternal age slope between fathers, and the influence of the environmental exposure(s). In addition, two scenarios for the induced DNM burden were modelled: (1) a constant number of DNM on top of the baseline mutational load; and (2) a model that simulates accumulating DNM with the duration of exposure. Below we outline how data were simulated and statistical power calculated.

2.1. Simulated data sets and associated assumptions

The mutation-by-paternal age rate is a key parameter that is required for simulating these data. Mutation-by-paternal age slope was set as 1.34 additional DNM transmitted to offspring per year by taking the weighted average across several published studies [10–12,28,29,36]. Maternal age-by-mutation slope is not considered here because: i) the maternal effect has a much smaller contribution than the paternal effect [12]; and ii) the maternal effect is partially accounted for due to the collinearity between the age of parents.

A recent study shows that as males age the rate at which they pass on DNM to their offspring is highly varied, with different fathers having statistically different slopes [10]. This may result from a combination of genetic and environmental factors. Therefore, we applied a simulation with a unique mutation-by-paternal age slope per family by sampling from a normal distribution with a mean of 1.34 and standard deviation of 0.8 [10]. These parameters on baseline mutation-by-paternal age slope and between-father variability of the paternal age effect were used throughout to generate data; however, other parameters were varied to either explore different scenarios or to investigate the sensitivity of power across effects sizes.

Finally, we varied the effect size of the mutagenic environmental exposure to evaluate the required sample to achieve 80% power across a range of potential effect sizes. While effect size is a valuable concept to describe the impact of a mutagen, it is problematic when comparing models. This is because different statistical tests use distinct formulas to calculate effect sizes, making it imprecise to compare statistical power between different models. Therefore, in this study, we use the term effect size when referring to general patterns, but we operationalise the effect sizes as either percent increase or slope change due to exposure to a mutagen for the constant and accumulating effect scenarios, respectively.

Using the above parameters, we simulated sample data sets in R (https://www.r-project.org/; see Supplementary Information) and verified that the simulated DNM for the control group fell in the appropriate range. The size of each data set was determined by the number of families being sampled, with half of these families being randomly assigned to the environmental exposure group and the other half to the unexposed control group. Each offspring had a paternal age selected from a uniform distribution between 18 and 37 years old. For multi-sibling families, the paternal age range was constrained to be a minimum of eight years apart between the youngest and oldest children, because there needs to be a modest age range to accurately estimate the slope of each family. Our parameter resulted in simulated data that provided a mean number of DNM per child of 60, which falls in the

Table 1
Comparison of three statistical models to detect an increase in *de novo* mutations.

	ANOVA	Multiple regression	Generalized Linear Mixed Models (GLMM)
Diagram			
Advantages	Parsimonious test. Easy to apply and interpret.	Able to model a more biologically realistic system (*e.g.*, controls for the known effect of paternal age), which increases the model's power. Tests if there is an additive or synergistic effect of multiple stressors on germ cell DNA damage, by including an interaction term. Additional predictor variable can be used to control for possible confounds (*e.g.*, socio-economic class).	Models family level variability as a random effect, which accounts for the variation in exposures and genetic effects on mutation-by-age slopes. This makes it the most realistic and powerful model. Models the statistical interaction. Able to detect an effect even when there is one family that is an outlier.
Disadvantages	Will only detect very strong effects. Cannot control for confounding variables.	Samples only one child per family and therefore ignores family-level variability.	Difficult to find cohorts with large pedigrees and sufficient sample size for control and exposed groups. Large families may be less representative of the general population.

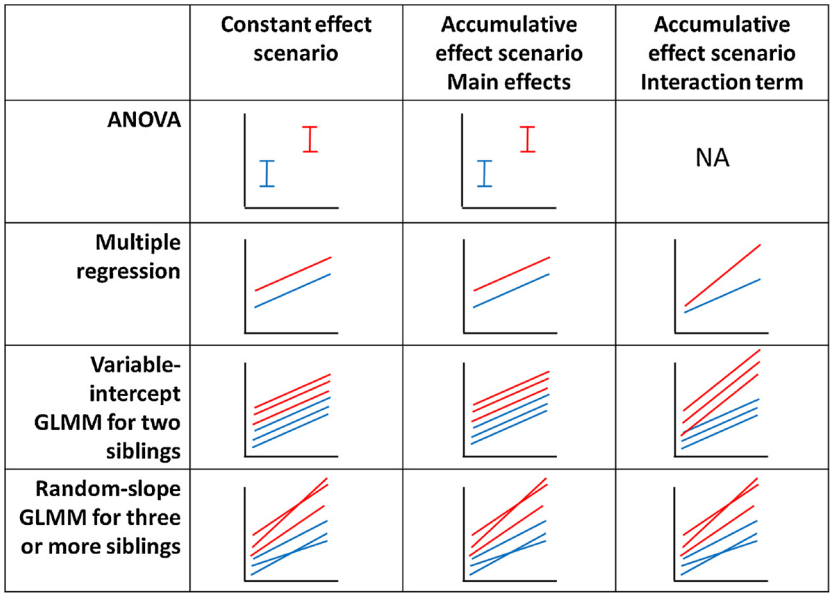

Fig. 1. Visual comparison between the statistical models (in the rows) and our scenarios (in the columns). NA represents a case where the statistical model is not possible for that scenario (*e.g.*, a one-way ANOVA cannot fit an interaction term).

expected range of DNM per offspring from the literature [10–12,28,29] for the control group. This confirmed that our distribution of simulated DNM corresponds to data from existing studies.

Each family was simulated to have a distinct mutation-by-paternal age slope. Within the exposed group, how these slopes were generated depended on whether a constant increase in DNM over paternal age or accumulating DNM over paternal age was specified (both described in the two scenarios below).

2.1.1. Scenario one – constant effect of mutagen exposure on DNM above the paternal age slope

The constant effect scenario assumes that the exposure's effects are not accumulating, or that the exposure was in the past, and therefore, a fixed additional number of DNM is added to each offspring regardless of the age of father at conception in the exposed group. Because the impact of environmental exposures on human germ cell mutations is unknown, there is uncertainty about the number of additional DNM that should be added to the offspring of the exposed group. We considered percent increases in the transmission of DNM above baseline from 10–40% as a possible range. This is a reasonable range based on, for example, a literature review on exposure to cigarette smoke and its heritable genetic effects [14]. Having assumed that humans have a mean of 60 DNM per offspring, then a 10% increase is equivalent to six additional DNM, 20% = 12 DNM, 30% = 18 DNM, and 40% = 24 DNM. The simulated DNM for each offspring were calculated based on paternal age at conception, mutation-by-paternal age rate of the family, and a fixed extra number of mutations if the father was a smoker, using a Poisson distribution. A Poisson distribution is appropriate as DNM is a count variable, and previous studies have established that DNM indeed follow a Poisson distribution [10,12].

2.1.2. Scenario two – accumulating effect of mutagen exposure on DNM above the paternal age slope

For the accumulating effect scenario, offspring of exposed fathers inherit increasing numbers of DNM above the paternal-age DNM, as would occur during chronic exposures with mutations accumulating in stem cells. This statistical interaction is depicted by a mutation-by-paternal age slope for the control group of 1.34 mutations per year, and a

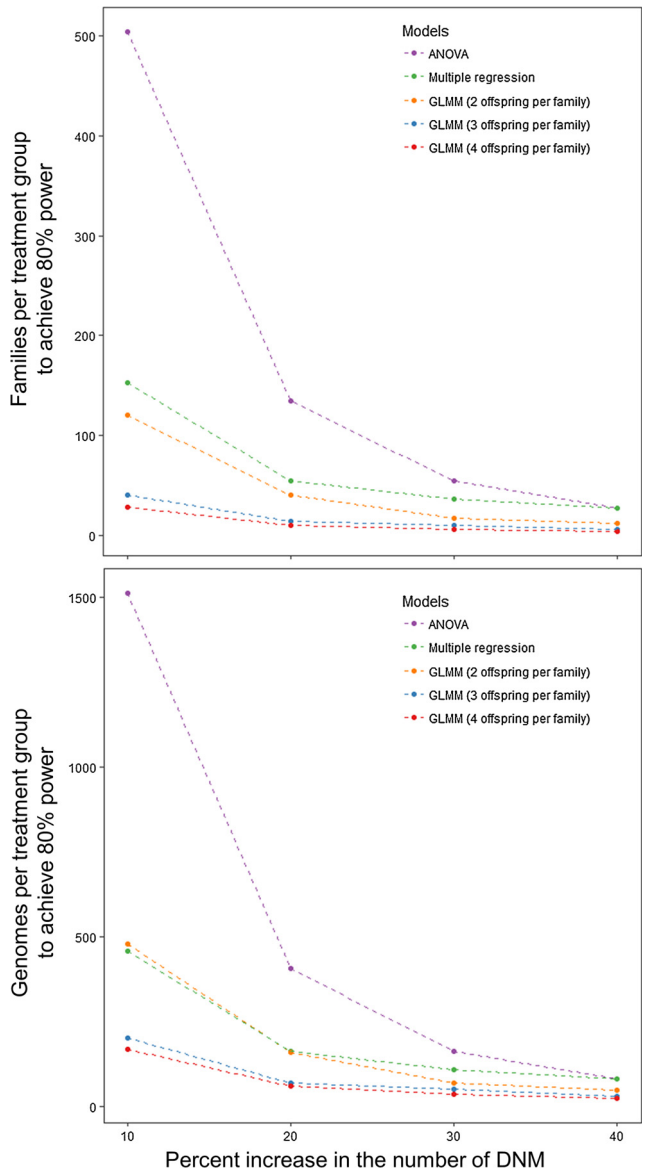

Fig. 2. a & b: Power curves for scenarios with a constant effect of increasing size. The x-axis shows the percent increase in mutational burden in offspring. Percent increase varies from 10%, 20%, 30%, and 40%, which is equivalent to an additional 6, 12, 18, and 24 extra DNM, respectively. The y-axis for Fig. 2a is the number of families per treatment group to achieve 80% power, and for Fig. 2b is the total genomes—parents and siblings—per treatment group to achieve 80% power. Each coloured line represents different fitted models and sampling strategies (ANOVA (purple), multiple regression (green), GLMM with two offspring per family (orange), GLMM with three offspring per family (blue), and GLMM with four offspring per family (red)).

steeper slope for the exposed group. Specifically, the magnitude of the effect was set to range from 0.2-0.8 mutations per year steeper than the baseline slope. For a weak 0.2 interaction effect, the paternal age effect would change from 1.34 to 1.54 DNM per year due to the added contribution of an environmental exposure. These parameters provide a similar amount of additional mutations to our constant model, but these DNM are redistributed, so the offspring that are born later received proportionally more exposure-induced DNM. This is biologically plausible as older fathers may accumulate more mutations due to a longer chronic exposure, or because of the combined stresses of mutagen exposure, age, and declining DNA repair capacity with age [37–39]. The simulated DNM were again drawn from a Poisson distribution, but with the adjustment to the paternal age effect slope for the exposed group.

2.2. Power analyses

The two scenarios above were simulated *in silico* to investigate the likelihood of detecting a difference in numbers of DNM in the offspring of control and exposed fathers. By repeating statistical tests one thousand times on different simulated data, the proportion of times the null hypothesis was rejected was counted to calculate statistical power.

We evaluated the power of different statistical models of increasing complexity: ANOVA and multiple regression sampled one-child per family, and General Linear Mixed Models (GLMM) sampled more than one child per family. ANOVA cannot control for the paternal age effect. The multiple regression included a paternal age effect with each father contributing only one data point. This multiple regression is constrained to sample only one child per father to ensure statistical independence; therefore, this model cannot control for inter-family variability. The GLMM controlled for inter-family variability, fitting a random effect to each father, allowing fathers to contribute data from more than one offspring. Here we assumed that there was modest inter-family variability, and then asked what was the improvement to power when modeling this extra parameter (*e.g.*, the family-level effect). The advantages and disadvantages of these three statistical approaches are provided in more detail in Table 1.

Statistical models were performed in the open source R programming language. The ANOVA model used the base R lm() function, while the multiple regression and GLMM were performed using the lme4() R package [40]. The multiple regression model is equivalent to an ANCOVA and the GLMM fits a random intercept or random slope mixed model. Below we provide a list of the variables used for these models, as well as the R code that was used to define these statistical models. Graphical representations of these models are shown in Fig. 1.

2.3. Variables

dnm = the number of DNM transmitted to an offspring (continuous response variable)

ageOfFather = paternal age of the father at conception of the offspring (continuous predictor variable)

exposure = whether the father of the offspring had been exposed to environmental agents prior to the conception of the offspring (binary predictor variable)

fatherID = the unique identifier of the father (categorical predictor variable)

2.4. Models and R code

ANOVA:
lm(dnm ~ exposure)
Multiple regression:
glm(dnm ~ ageOfFather + exposure, family = poisson)
Random-intercept GLMM:
lmer(dnm ~ ageOfFather + exposure + (1 |fatherID), family = poisson)
Random-slope GLMM:
lmer(dnm ~ ageOfFather + exposure + (1 + ageOfFather|fatherID), family = poisson).

The above models are for the constant effect scenario. These models can detect if there is a difference in mutational burden (*e.g.*, a main effect of the exposure that leads to an overall change in the number of DNM in exposed versus controls). The below models are modified to investigate an accumulating effect scenario. However, under the accumulating scenario, an additional analysis can be applied to assess whether there is a change in the slope of exposed relative to control fathers (*i.e.*, a statistical interaction). Three additional models are provided that were used to model the interaction term (not merely test for a difference).

Table 2
Comparison of the sample sizes required for the constant effect scenarios across four statistical models.

Models	% increase in mutations	# families per group	# genomes per group[a]
ANOVA	10%	504	1512
ANOVA	20%	135	405
ANOVA	30%	54	162
ANOVA	40%	27	81
Multiple regression	10%	153	459
Multiple regression	20%	54	162
Multiple regression	30%	36	108
Multiple regression	40%	27	81
GLMM (2 offspring per family)	10%	120	480
GLMM (2 offspring per family)	20%	40	160
GLMM (2 offspring per family)	30%	17	68
GLMM (2 offspring per family)	40%	12	48
GLMM (3 offspring per family)	10%	40	200
GLMM (3 offspring per family)	20%	14	70
GLMM (3 offspring per family)	30%	10	50
GLMM (3 offspring per family)	40%	6	30
GLMM (4 offspring per family)	10%	28	168
GLMM (4 offspring per family)	20%	10	60
GLMM (4 offspring per family)	30%	6	36
GLMM (4 offspring per family)	40%	4	24

[a] Number of genomes per group is the total number of parents and siblings required to achieve 80 % power, per group.

Multiple regression with interaction term:

glm(dnm ~ ageOfFather + exposure + ageOfFather* exposure, family = poisson).

Random-intercept GLMM with interaction term:

glmer(dnm ~ ageOfFather + exposure + ageOfFather* exposure + (1|fatherID), family = poisson)).

Random-slope GLMM with interaction term:

glmer(dnm ~ ageOfFather + exposure + ageOfFather* exposure + (1 + ageOfFather|fatherID), family = poisson)).

All of the above statistical models make different assumptions about the biological units of replication and require different sampling designs. The ANOVA/multiple regression models consider that each offspring is derived from an independent meiosis, and assume that there are no family-level differences; specifically, the multiple regression model—with its associated assumptions—includes the main effect of paternal age, but does not include an additional parameter to model family variability. Although ignoring the contributions of genetics and possible exposures to the rate of DNM within each family may dilute statistical power, large families may not always be available for analyses. However, when possible, a family-level variable can be included in the GLMM. For the GLMM, there needs to be at least two offspring per family in order to fit a line for each family. The choice between these statistical models is equivalent to a sampling choice. For the same reasons, the ANOVA/multiple regression can sample many families each with only one offspring, while the GLMM can sample fewer large families with at least two offspring. Overall, in this analysis we investigated whether there is more power in sampling one child per family across many families, versus sampling many siblings across fewer families (while allowing us to control for between-family differences).

3. Results

3.1. Scenario one – constant effect of mutagen exposure on DNM above the paternal age slope

The power analyses showed that there were differences in the required sample sizes between our five modeling/sampling approaches for the constant effect scenario (Fig. 2a & b; Table 2). The ANOVA under-performed and required a larger sample size than all other models. For example, 504, 135, 54 and 27 families per group were needed to obtain sufficient power to detect a 10%, 20%, 30% and 40% increase in DNM, respectively, using the ANOVA. Conversely, 153, 54, 36 and 27 families were necessary when using multiple regressions that controlled for paternal age. As the effect size shrinks the benefit to modeling paternal age increases.

The addition of a parameter to model inter-family variability, through sampling more siblings, further improved power. The paternal age effect was modelled by both the multiple regression and the three GLMM, but only the GLMM model fits unique paternal age effects for each family. Consistently fewer families per treatment group were needed to achieve sufficient power with the GLMM approach (Fig. 2a & b; Table 2). For the largest effect size, the multiple regression required 27 families per group, whereas with the four sibling GLMM this was reduced to four families (*i.e.*, an 85% reduction in sample size). For the smallest effect size, the multiple regression required 153 families per group, while with the four sibling GLMM this was reduced to 28 families. As within family sampling increased from two, three, and four siblings per family, statistical power increased. The GLMM with two siblings per family was intermediate to the multiple regression model and the GLMM with three siblings per family (Fig. 2a). Sampling more siblings within a family allowed for a reduction of total genomes needed to achieve 80% power using the random slope models (*i.e.*, GLMM with three or four siblings), but not for the random-intercept model with two siblings per family (Fig. 2b). Between the three and four sibling random slope GLMM, sampling an additional sibling achieved sufficient power with fewer families (Fig. 2a) and genomes (Fig. 2b) per group; for the largest effect size the GLMM with three siblings required six families per group, while with the GLMM with four siblings required only four families.

3.2. Scenario two – accumulating effects of mutagen exposure on DNM above the paternal age slope

3.2.1. Main effect for scenario two – increase in total number of DNM in exposed versus control fathers

The accumulating effect scenario power analysis showed that GLMM achieved higher power and reduced the effective samples size required to detect an increase in the number of DNM resulting from an exposure (Fig. 3, Table 3). Each statistical model differed markedly in their required samples sizes (Fig. 3; Table 3). The ANOVA model required the most samples regardless of the slope increase above baseline for the exposed group. For the smallest effect size (*e.g.*, a slope increase of 0.2 additional mutations per year for the exposed group), 630 versus 59 families per group were required for the ANOVA and four sibling

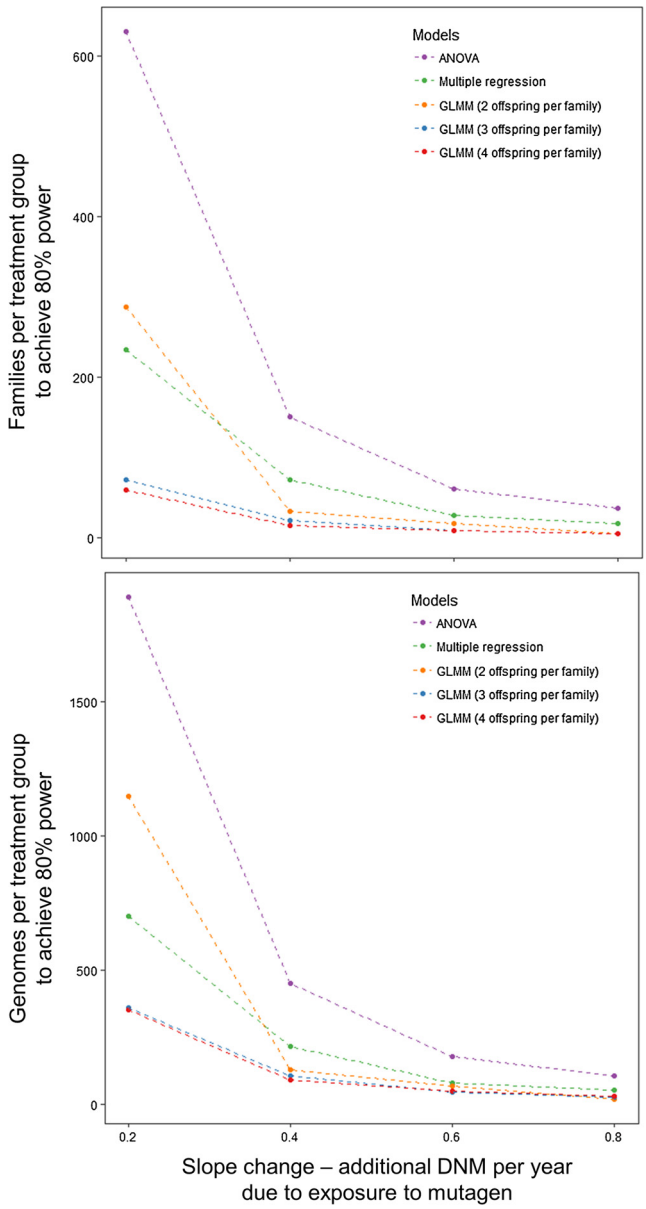

Fig. 3. a& b: Power curves for detecting a difference in the number of DNM between control and exposed groups (*e.g.*, a main effect) for the accumulating mutations scenario across a range of effects. The x-axis represents different scenarios where the interaction effect becomes more pronounced; that is, the slope increases between the control and exposed groups from near parallel to a steeper gradient. The y-axis for Fig. 3a is the number of families per treatment group to achieve 80% power and for Fig. 3b is the total genomes—parents and siblings—per treatment group to achieve 80% power. Each coloured line represents different fitted models and sampling strategies (ANOVA (purple), multiple regression (green), GLMM with two offspring per family (orange), GLMM with three offspring per family (blue), and GLMM with four offspring per family (red)).

GLMM, respectively (*i.e.*, a 91% reduction in families sampled). For the largest effect size, the ANOVA required 36 families per group, while the four sibling GLMM required five families (*i.e.*, an 86% reduction).

A similar result was obtained when comparing the multiple regression with the GLMM models. For the largest effect size, the multiple regression required 18 families per group versus five families (*i.e.*, a 72% reduction) with the four sibling GLMM. For the smallest effect size, the multiple regression required 236 families per group, whereas only 59 families were needed using the four sibling GLMM (*i.e.*, a 75% reduction). There was one exception where the GLMM required a larger sample size than the multiple regression model; for the smallest effect size, the multiple regression required 234 families per group, but the two sibling variable-intercept GLMM required 287 families per group.

Next, we compared the performance of the GLMM with three and four siblings and found that sampling the larger multi-sibling families provided a benefit only for small effect sizes. For the smallest effect size (a slope change of 0.2 additional mutations per year), the three and four sibling GLMM required 72 and 59 families per group (*i.e.*, a 18% reduction in sampling effort for the four sibling GLMM), respectively. The benefit of the four sibling GLMM was diminished for the largest effect size (a slope change of 0.8 additional mutations per year), with both models requiring five families per group (Fig. 3b; Table 3) to reach 80% power; therefore, for this scenario, sampling three siblings per family achieves the same power of sampling four siblings per family and requires the sequencing of five fewer genomes (Fig. 3b; Table 3).

3.2.2. *Interaction term for scenario two: increase in the age-mutation slopes of the exposed fathers relative to controls*

Roughly three times more families were required to detect an interaction term than a main effect. For the accumulating effect scenario with the largest effect size, the GLMM with four siblings required five and 14 families per group to detect a main and interaction effect, respectively (Table 3). For the same situation, the multiple regression models required 18 and 69 families per group to detect a main and interaction effect (Table 3), respectively. This comparison could not be made for the ANOVA, as our one-way ANOVA model cannot fit an interaction term (see Fig. 1).

Aside from the much larger sample sizes required for detecting interactions, general patterns in the effective sample sizes were consistent with the analyses above. For example, the GLMM required fewer samples than the multiple regression models. For the largest to smallest effect size, the multiple regression models required between 69–828 families per group, respectively. In contrast, the four sibling GLMM required between 14–97 families per group. The multiple regression required more families per group than both the random-intercept and slope GLMM across a range of effect sizes (Fig. 4a; Table 3). Furthermore, as the effect size decreased, the multiple regression required proportionately more samples than the GLMM (Fig. 4).

As for the constant scenario, the difference in required sample size was much narrower when comparing the random slope GLMM with three and four siblings, and in some situations the benefit of the larger multi-sibling family disappeared. For the weakest interactive effect (0.2 mutations per year beyond the baseline rate), sufficient power was obtained with 160 and 97 families per group, with three and four siblings respectively (*i.e.*, a 39% reduction in families). The advantage of the GLMM with four siblings diminished as the effect size became more prominent (*e.g.*, as the slopes between the control and exposed group become increasingly divergent). For the steepest interaction effect of 0.8 additional DNM per year, both GLMM with three and four siblings per family achieved 80% power with 14 families per group (Fig. 4a). For the edge case where slope change is 0.8, these conditions allow sampling three siblings per family to achieve the same power as that achieved by sampling four siblings per family (*e.g.*, see the slight crossing of the red and blue lines on the far right of Fig. 4b).

4. Discussion

There is much interest in the application of WGS to detect environmental exposures that can cause heritable genetic effects [14,21]. Our simulation study demonstrates that exposure-related increases in germline mutation rates can be more readily detected when the mutation-by-paternal age effect is modelled. Specifically, ANOVA models that do not include age of father as a variable require much larger samples sizes to achieve sufficient power; compared to the multiple regression and GLMM with four siblings, the ANOVA model requires up to an additional 351 and 476 families per group, respectively. In

Table 3
Comparison of the sample size required for the accumulating mutations scenario to detect either the main effect of exposure or an interaction term between paternal age effect and exposure.

Models	Main effects			Interaction term	
	Slope increase above baseline	# families per group	# genomes per group	# families per group	# genomes per group
ANOVA	0.2	630	1890	NA	NA
ANOVA	0.4	150	450	NA	NA
ANOVA	0.6	60	180	NA	NA
ANOVA	0.8	36	108	NA	NA
Multiple regression	0.2	234	702	828	2484
Multiple regression	0.4	72	216	216	648
Multiple regression	0.6	27	81	111	333
Multiple regression	0.8	18	54	69	207
GLMM (2 offspring per family)	0.2	287	1148	494	1976
GLMM (2 offspring per family)	0.4	32	128	143	572
GLMM (2 offspring per family)	0.6	17	68	76	304
GLMM (2 offspring per family)	0.8	5	20	42	168
GLMM (3 offspring per family)	0.2	72	360	160	800
GLMM (3 offspring per family)	0.4	21	105	71	355
GLMM (3 offspring per family)	0.6	9	45	38	190
GLMM (3 offspring per family)	0.8	5	25	14	70
GLMM (4 offspring per family)	0.2	59	354	97	582
GLMM (4 offspring per family)	0.4	15	90	38	228
GLMM (4 offspring per family)	0.6	8	48	22	132
GLMM (4 offspring per family)	0.8	5	30	14	84

*Number of genomes per group is the total number of parents and siblings required to achieve 80 % power, per group.

relative terms, this is a 94% increase in sample size to detect a constant effect scenario with six induced mutations compared to the best performing model. Therefore, it is important to use statistical models that control for paternal age when investigating the effects of potential human germ cell mutagens on the rate of DNM in the offspring.

Although accounting for paternal age effect provides a large improvement in power, we found that modeling family-to-family variability further reduces the required sample size, especially when the effect size is small. Using families with ≥ two siblings, the GLMM is able to fit a line to each family, allowing the model to account for differences in mutation-by-paternal age rates, which is highly varied [10]. The GLMM also provides an improvement in power from the multiple regression model, in particular for small effect sizes. For a constant effect scenario with the weakest effect size (six additional mutations per year), the required sample size is 153 and 40 for the multiple regression and GLMM with three siblings, respectively; this equates to a 74% reduction in required sample size for the GLMM with three siblings. An incremental improvement also occurs when the model changes from random-intercept to random-slope. In general, GLMM requires fewer families per group, and this is consistent for both constant and accumulating effect scenarios.

Applying the GLMM model to families with four offspring allows a more accurate estimation of slopes, further reducing required sample sizes. If the exposure contributes an additional six mutations, there is a reduction in required sampled genomes by 32 and six for the constant and accumulating effects scenarios, respectively, compared with sampling families with three children. However, this advantage dwindles for the largest effect size; the same numbers of three- and four-child families are needed to achieve an 80% power for the largest effect size (Fig. 4b). Overall, combining the benefits of mixed effect models to control for family-to-family variability, and sampling fewer larger multi-sibling families, provides the opportunity to detect increases in DNM with fewer human genome sequences.

Our results show that larger sample sizes are required when there is an accumulative effect between paternal age and exposure to mutagens compared with a constant effect. The exposure is being modelled with a difference test for the constant scenario and a slope change for the accumulating scenario. Not surprisingly, entirely different models provide a modest difference in recommended samples sizes. Our power analysis reveals that to detect a main effect resulting in an additional 24 DNM (*e.g.*, either a 40% increase or a slope change of 0.8 additional mutations per year) using a GLMM for constant and accumulating scenarios, at least four and five families per group should be sampled, respectively. However, a much larger number of families are required to detect weaker accumulating mutagenic effects. To detect the interaction term (not merely a difference in mutational load) requires even larger sample sizes.

Sampling two siblings per family provides power that is generally intermediate between the multiple regression and the three-child random-slope GLMM, with the advantage that these smaller families are more common in many cohorts. The two siblings per family GLMM model applies a random-intercept model. It increases the sample size by going from a single to two offspring, and captures a portion of the family variability from this additional intra-family sample. Our simulated data set assumes that families vary in their mutation-by-paternal age slopes (as in [10]). Consequently, models that only allow the intercept to vary provide some additional power by better accounting for family variation, but are unable to account for all of the family variation. If the cohort has insufficient large multi-sibling families, but has an expected large effect size, then sampling two siblings achieves consistently the highest power with respect to the multiple regression model. Assuming the constant effect scenario, with 40% more induced DNM, then the two-sibling random-intercept model requires 12 families per group, while the multiple regression requires 27. Its relative performance worsens as the effect size decreases (Fig. 2b & b), with very minimal benefits to sequencing the extra child's genome at the smallest effect size. Overall, if the cohort has insufficient large multi-sibling families, and there is an expected weak effect size, the best design may switch from families with two siblings to sampling families with single offspring (applying a multiple regression model).

The performance of the random intercept model improves when an interaction term is included (Fig. 4b); this is not surprising as the inclusion of an interaction term provides the model with some flexibility to fit different slopes between the control and exposed groups, achieving performance that is intermediate between the multiple regression and the three sibling GLMM. However, as with all power simulation studies, this result should be interpreted with caution as the initial parameters influence the modeling recommendations.

Our analysis makes several assumptions (*e.g.*, the degree of variability between families and the magnitude of the expected effect), and

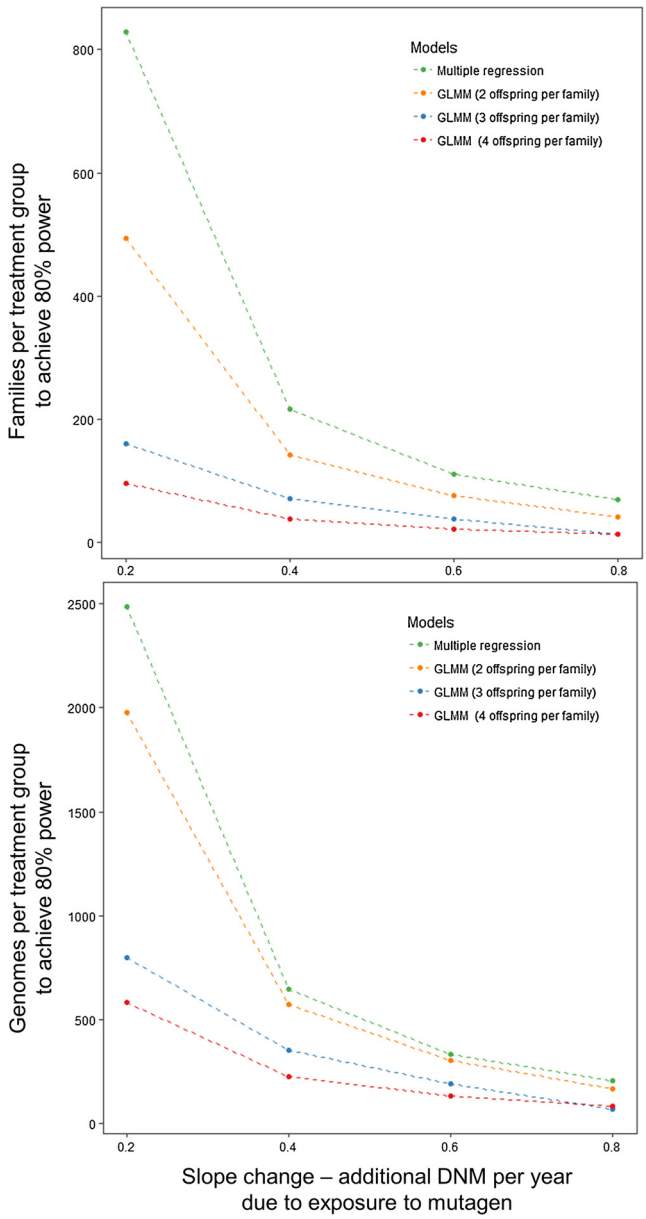

Fig. 4. a & b: Power curves for detecting an interaction term between mutation-by-paternal age and exposure treatments. The x-axis represents different magnitudes of the interaction effect, from near parallel to a steeper gradient. The y-axis for Fig. 4a is the number of families per treatment group to achieve 80% power and for Fig. 4b is the total genomes—parents and siblings—per treatment group to achieve 80% power. Each coloured line represents different fitted models and sampling strategies (multiple regression (green), GLMM with two offspring per family (orange), GLMM with three offspring per family (blue), and GLMM with four offspring per family (red)).

changes to these assumptions may modify our conclusions. The relative performance between the GLMMs will depend on how—and to what extent—families vary in their mutation-by-paternal age slopes. The only study on this subject compared three families and found a significant main effect between mutation-by-paternal age rates [10]. This simulation study uses a recent best estimate of between-family variability, but with only three families to base this on, this estimate may be subject to change. If the family variability parameter is an overestimate, there will be smaller sample size differences between the multiple regression and GLMM; alternatively, if the family variability parameter is an underestimate, there will be larger sample size differences between the multiple regression and GLMM. Further work is needed to provide a more accurate estimate of between family variability to increase the confidence in selecting statistical models.

These sample size recommendations are based on detecting a difference in the number of DNM. In addition, WGS allows for complementary mechanistic predictions to be tested that can increase confidence in a causal relationship. For example, the data could be used to determine if the additional DNM in the exposed group correspond to a change in the underlying mutational processes consistent with anticipated changes in mutation spectrum. Evidence for a spectral shift could provide compelling causal evidence that small statistically significant changes in mutation rate are biologically relevant. Further work is needed to determine if our sample size recommendations are sufficient to detect shifts in the mutational signatures.

This work shows that the identification of a human germ cell mutagen using WGS is feasible with existing sequencing platforms and well-chosen statistical models, if the right cohort is available. For example, sequencing six families per treatment group would be sufficient to achieve 80% power to detect a 30% increase in DNM using 4-child families. As technologies improve, sequencing accuracy will increase, allowing the identification of significant effects in even smaller sample sizes; existing sequencing platforms and algorithms are currently under calling DNM (*e.g.*, there is a high sensitivity but a lower specificity), with 1/5 of the genome being inaccessible to DNM discovery due to technical issues. The major challenge at present is the identification of appropriate cohorts where environmental exposures are fully characterised, with appropriate controls for confounding factors. Detection of small increases in DNM have far-reaching impacts on a global population scale [14]; thus, our study provides compelling support that such studies should be undertaken.

In conclusion, our data demonstrate that mixed models, which can control for paternal age, and sampling of large multi-sibling families, achieve the highest power to detect an increase in DNM. Pragmatic study design decisions for the detection of human germ cell mutagens depend on the attributes of cohorts and the effect size of the mutagen. If the cohort has sufficient large multi-sibling families (*e.g.*, exceeding the minimal number of families per group from Tables 2 & 3), then sampling three or more offspring will achieve the highest power. This approach is resilient to detecting germ cell mutagens even when there is high between-family variability. If the cohort has insufficient large multi-sibling families, then sampling two siblings becomes the highest power test for most conditions. Furthermore, a mixed model with multi-sibling families requires fewer total genomes to be sequenced, making this analytical strategy the most cost-effective approach.

Competing interests

None.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.mrgentox.2018.04.004.

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